# Aditya Jha

aditya.jha2020123@gmail.com | github.com/AJ576 | linkedin.com/in/aditya-jha777

## Education

**The City College of New York** — New York, NY
*Bachelor of Science in Computer Science* — *Graduation: Spring 2027*
- GPA: 3.79 | Relevant Coursework: Algorithms, Operating Systems, Distributed Systems, Computer Security

## Experience

**Bloomberg** — Jun 2026 - Aug 2026
*Software Engineering Intern* — *New York, NY*
- Built a **C++ real-time market-data ingestion service** from scratch, normalizing JSON WebSocket feeds from **5 exchange vendors** into a common format and delivering ticks into **Bloomberg's internal market-data pipeline**
- Designed a **single-threaded, event-driven architecture** to minimize per-message latency, scaling horizontally across processes; **configuration-driven parser** handled up to **10 concurrent channels** per instance and onboarded new vendor formats without core code changes
- Built a connection lifecycle **state machine** (authentication, subscriptions, reconnection with **exponential backoff**) to keep feeds running through vendor disconnects; added **unit tests** and **observer-based monitoring/alerting**

**Break Through Tech x Cadence** — Sep 2025 - Dec 2025
*Tech Fellow* — *New York, NY*
- Built a **hybrid resume-to-job matching system** in a team of 6, combining **e5-large embeddings** with a **gradient-boosted model** to rank **2.4K resumes against 5.4K job postings** through a **FastAPI** backend
- Distilled a **335M-parameter embedding model** into an interpretable **5-feature gradient boosting model** trained on **115K pseudo-labeled pairs**, reaching **0.86 ROC-AUC** and **0.90 Recall@10** on held-out jobs

**Unadat** — Jul 2025 - Aug 2025
*Software Engineering Intern* — *New York, NY*
- Redesigned a task-management UI in **JavaScript** on top of optimized **PHP backend APIs** for **100+ active users**, and built a terminology wrapper system that extended the platform from **B2C** into **B2B** markets

## Projects

**EquiLearn** — Swift, SwiftUI, Core Haptics, AVFoundation — Github
- **Co-first-author**, **published at MIT URTC 2026**: iOS app (2-person team) that lets **blind and low-vision students** explore math graphs by touch with real-time **haptics, pitch/pan sonification, and VoiceOver**, plus an AI tutor; iterated across multiple sessions with BLV testers and an NYPL assistive-technology coordinator
- Wrote the **math engine from scratch** (tokenizer, recursive-descent parser, cached domain-safe AST evaluator, bisection root-finding) and a **real-time audio synth** on an AVAudioSourceNode render thread; fixed a sample-rate bug that made **every tone ∼9% sharp**

**Galactic Cruise** — Three.js, WebGL/GLSL, Vite — Github | Demo
- Built a procedural space-exploration game with seeded solar systems, **quadtree-LOD terrain streamed by a Web Worker pool** down to ∼4 m vertex spacing, Keplerian orbits, and **analytical eclipse shadows in GLSL** (no shadow maps); **170+ automated tests**
- Implemented a **6DOF flight model** with swept-sphere terrain collision, first-person walking/swimming, a **4-phase hyperspace state machine**, and **instanced asteroid belts** streamed in deterministic spatial cells; **logarithmic depth buffer** spans 40 cm to 400,000 km

**Cosmere Archivist (RAG)** — Python, FastAPI, Next.js, FAISS, Docker — Github | Demo
- Built and deployed a **RAG Q&A app** over **4,740 wiki articles (19.8K passages)**; fit the backend into a **512 MB container** by memory-mapping the FAISS index and using a byte-offset metadata store, **cutting metadata heap 97% (25 MB → 0.7 MB)**
- Built the **full pipeline**: scrape, header-aware chunking, dedupe, MiniLM embeddings, **FastAPI** (`/ask`, `/search`), and a **Next.js** client on Vercel with retry logic for free-tier cold starts; **Dockerized** (CPU-only PyTorch, model baked in at build) with **20+ automated tests**

## Technical Skills

**Languages:** C++17, Python, Swift, JavaScript, SQL
**Frameworks & Libraries:** SwiftUI, React, Next.js, FastAPI, Three.js
**Databases:** PostgreSQL, MySQL, Firebase
**Tools & Systems:** GCP, Docker, Linux, POSIX, CMake, Git, CI/CD

## Leadership & Awards

**Leadership:** Founder & VP, Quant Finance Club | Sponsorship Chair, Google Developer Group (GDG)
**Hackathons:** 1st BYTE Hacks | 2nd Hack Knight | 3rd BlackStone Innovation Sprint